

Butane dihedral barrier crossing in water: reactive flux vs. memory kernel + anharmonicity

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1 Question

The Grote–Hynes (GH) transmission coefficient κ_{GH} is computed from the extracted memory kernel $K(t)$ via the implicit equation

$$\lambda_r = \frac{\omega_b^2}{\lambda_r + \widehat{K}(\lambda_r)}, \quad \kappa_{\text{GH}} = \lambda_r / \omega_b, \quad (1)$$

which assumes a *parabolic* barrier of curvature ω_b^2 . For butane dihedral in water, kernel-based GH gives $\kappa_{\text{GH}} \approx 0.89$, yet the direct reactive-flux measurement gives $\kappa_{\text{RF}} \approx 0.25$ — a factor of ~ 3.5 disagreement. We test the hypothesis that the gap is due to *anharmonicity* of the dihedral PMF away from ϕ_b (rather than a kernel-extraction problem) by running a 1D generalized Langevin equation (GLE) with the same extracted $K(t)$ and the *full* mean-force spline, and measuring its reactive flux directly.

2 System and shooting protocol

GROMOS 53A6 united-atom butane in 722 SPC/E waters (GROMACS 2020.7 with PLUMED 2.9 for the dihedral CV). NPT, $T = 300$ K, $P = 1$ atm, all-angle constraints. Reaction coordinate is the dihedral ϕ . Working with the *eclipsed* barrier at $\phi_b \approx 120^\circ$ (PLUMED) $\equiv 245^\circ$ (numpy convention), between the trans and gauche⁺ wells.

Phase A — bath equilibration. Single 1500 ps trajectory with PLUMED RESTRAINT KAPPA=10000 at ϕ_b , warm-up 500 ps, then 51 bath snapshots saved every 20 ps ($\rightarrow$ `shooting_barrier/equil/config_*.gro`).

Phase B — shooting. For each seed $i = 1 \dots N_{\text{shoot}}$, free dynamics of 10 ps (5001 frames at $\Delta t = 2$ fs), `gen-vel=yes`, `gen-seed=i`, no PLUMED restraint. Each seed picks one of the 51 bath configs cyclically. Recorded per frame: $\phi(t)$, Jv (generalized velocity through the dihedral Jacobian), and $\dot{\phi}$ via central FD. Saved as `shoot_seed{i}/phi_data.npz`. Currently $N = 1006$ trajectories with valid `phi_data.npz` (seed 807 orphaned: TRR present, post-processing interrupted). Of these, $N_{\text{accept}} = 663$ have $|\phi(0) - \phi_b| < 10^\circ$ and are used as “barrier-top” trajectories.

Code. `gromacs_run/shooting_barrier/run_shooting_barrier.py`.

3 Kernel extraction

Two extractors are available, both starting from the cold-start out-of-equilibrium least-squares formulation (`kernel_extraction/nonstationary_kernel_lsq.py`):

Free-LSQ: `extract_kernel_lsq` returns the kernel sampled on the discrete time grid $\Delta t_e = 5\Delta t = 10$ fs as a free n -vector. Tikhonov-smoothness regularized; here truncated to 500 fs.

Prony NLSQ: `extract_kernel_prony` parameterises $K(t) = \sum_{k=1}^{n_{\text{exp}}} a_k \exp(-t/\tau_k)$ and fits (a_k, τ_k) by nonlinear least squares against the same Toeplitz system. We added a `tau_bounds=( $\tau_{\min}, \tau_{\max}$ )` option to prevent runaway slow tails at large N .

Both use the binned mean force $\langle \ddot{\phi} | \phi \rangle$ as the deterministic part. The barrier curvature $\omega_b = \sqrt{|U''(\phi_b)|} = 24.29 \text{ rad/ps}$ is read from the spline second derivative at ϕ_b .

4 GLE simulation of $\kappa(t)$

For a given $K(t)$ and mean force $F(\phi)$, the GLE

$$\ddot{\phi}(t) = F(\phi(t)) - \int_0^t K(t-s) \dot{\phi}(s) ds + \eta(t), \quad \langle \eta(t) \eta(t') \rangle = \beta_{\text{eff}} K(|t-t'|), \quad (2)$$

is integrated numerically. Two integrators are implemented in `kernel_extraction/benchmark_extended_cases.py`.

1. `generate_gle_multiexp`: one Mori auxiliary variable per exponential of a Prony kernel. OU noise with FDT, exact in continuous time; numerically explicit (Euler–Maruyama) on the auxiliaries.
2. `generate_gle_arbitrary`: arbitrary numeric $K(t)$. Friction term computed as $\Delta t \sum_j K_j v_{i-j}$ (rectangle convolution). Colored noise via spectral filter: $\eta = \text{IFFT}(\sqrt{\mathcal{R}\tilde{C}} \text{FFT}(w))$ where w is unit-variance white noise and $\tilde{C} = \text{FFT}(\beta_{\text{eff}} K)$ is the discrete spectral density of the autocovariance. The $\sqrt{S}$ scaling (without the spurious Δt factor present in the original code) reproduces $\langle \eta_i \eta_j \rangle = \beta_{\text{eff}} K(|i-j| \Delta t)$ exactly for circulant covariances.

The effective inverse temperature is read from the data: $\beta_{\text{eff}} = \langle \dot{\phi}(0)^2 \rangle = 30.29 \text{ rad}^2/\text{ps}^2$. Initial conditions: $\phi(0) = \phi_b$, $\dot{\phi}(0) \sim \mathcal{N}(0, \sqrt{\beta_{\text{eff}}})$, auxiliaries from their stationary distribution.

The reactive flux estimator

$$\kappa(t) = \frac{2 \langle \dot{\phi}(0) \theta(\phi(t) - \phi_b) \rangle}{\langle |\dot{\phi}(0)| \rangle} \quad (3)$$

is then computed both from the MD shoots (gives κ_{RF}) and from the GLE shoots (gives κ_{GLE}). For the GLE the run is $T = 5 \text{ ps}$, $\Delta t_{\text{GLE}} = 1 \text{ fs}$, $N_{\text{traj}} = 5000$.

5 Results

5.1 Headline numbers ($N = 663$, $\text{nbins}=36$, `polyfit` ω_b)

Method	κ plateau	Comment
κ_{RF} (MD shoots)	0.250	reactive flux, 3–8 ps plateau
κ_{GH} (Free-LSQ kernel)	0.899	analytic, eq. (1)
$\kappa_{\text{GLE}}^{\text{harm}}$	0.889	parabolic F , sanity check
$\kappa_{\text{GLE}}^{\text{anharm}}$	0.241/0.223	full $F(\phi)$ spline (stride=5 / 1)

The harmonic-barrier GLE reproduces the analytic κ_{GH} within 1%, validating the integrator and FDT noise scaling. Replacing the parabolic F with the full anharmonic spline drops κ from 0.90 to ~ 0.23 — recovering $\kappa_{\text{RF}} = 0.25$ within ~ 5 – 10% . The two anharmonic values bracket the result depending on the kernel sampling stride: stride=5 ($\Delta t_e = 10 \text{ fs}$) is the value used throughout the convergence figure (Fig. 1); stride=1 ($\Delta t_e = 2 \text{ fs}$) gives a slightly lower $K(0)$ and a slightly more dissipative effective kernel.

ω_b extraction. The barrier curvature is obtained by a parabolic polyfit of the free-energy profile (from cumulative integration of the binned $\langle \ddot{\phi} | \phi \rangle$) on a narrow window $|\phi - \phi_b| < 0.15$ rad, giving $\omega_b = 24.2$ rad/ps. This polyfit estimate is stable to the bin choice (24.0–24.5 for nbins $\in \{24, 36, 48\}$, only the 60-bin case dips to 20.6 from a local PMF artifact). The naive second-derivative-of-the-spline at ϕ_b is much less robust: it varies from 3.3 to 25.9 across the same range of bin counts, and was the source of an earlier inflation of κ_{GH} .

Stride sensitivity. Subsampling the trajectories before the kernel extraction (stride 1, 2, 5, 10 corresponding to $\Delta t_e = 2, 4, 10, 20$ fs) shifts $K(0)$ by up to 25 % but the integrated friction $\gamma_{\text{int}} = \int K dt$ only by ~ 8 %. Consequently κ_{GLE} moves by no more than ± 0.04 across this range, which is within the $\kappa_{\text{GLE}}/\kappa_{\text{RF}}$ scatter at $N = 663$.

$\phi(0)$ centering. The 663 accepted barrier-top trajectories have $\phi(0) = 244.99^\circ$ (target $\phi_b = 245.00^\circ$), $\sigma_{\phi(0)} = 3.14^\circ$, range $[236.3, 250.2]^\circ$ — i.e. the shoots are tightly localised at the barrier, so any visual parabola misfit in earlier figures is a curvature-extraction issue, not a shooting-protocol issue.

5.2 Convergence with sample size

The script `butane_kappa_convergence.py` repeats the comparison on bootstrap subsamples $N \in \{50, 100, 150, 200, 300, 500, 663\}$, recomputing κ_{RF} on the same subsample as the kernel fit. Both estimators converge to the same plateau as N grows; at $N \sim 100$ they already agree to within ~ 10 % but each is still noisy on its own. The full $N = 663$ result is reliable. See Fig. 1.

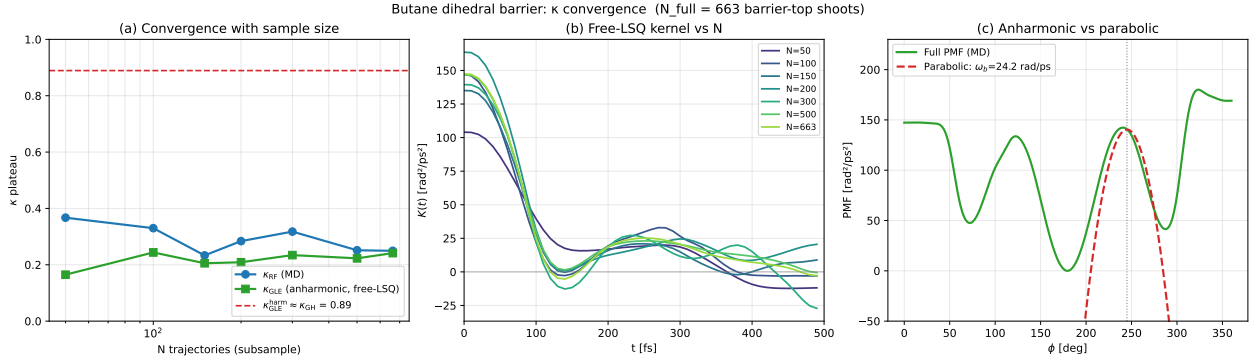


Figure 1: (a) κ_{RF} (MD) and $\kappa_{\text{GLE}}^{\text{anharmonic}}$ (GLE with full mean force) as functions of the sample size N used for both estimators. The dashed line shows $\kappa_{\text{GLE}}^{\text{anharmonic}} \approx \kappa_{\text{GH}}$ at full N . (b) Free-LSQ kernel for the same subsamples. (c) Full PMF (from binned $\langle \ddot{\phi} | \phi \rangle$) compared to the parabolic approximation at ϕ_b .

5.3 Why Prony alone is misleading

A 2-exponential Prony fit on the full $N = 663$ dataset returns one fast component $(a, \tau) \approx (248, 40$ fs) matching the 1-exponential fit, plus a *second* slow component (13, 10 ps) whose integrated friction (130 rad/ps) is an order of magnitude larger than the fast one but contributes negligibly to $\hat{K}(\lambda_r)$ at the GH solution ($\lambda_r \approx 6$ rad/ps). It is, however, dynamically active in the GLE simulation, where it depresses κ_{GLE} to ≈ 0.32 instead of 0.24. Switching to the Free-LSQ numeric kernel (truncated at 500 fs) avoids this artifact entirely.

6 Code map

```

gromacs_run/shooting_barrier/
  run_shooting_barrier.py      # GROMACS shooting from phi_b ~ 120 deg
  equil/                      # 51 bath configs
  shoot_seed{1..1007}/        # phi_data.npz per seed (1006 valid)
testASE/butane_dihedral_water/
  butane_rf_vs_kernel.py      # RF vs kernel-derived GH kappa; t0 sweep
  butane_gle_kappa.py         # GLE kappa simulation:
                              #   --method {prony2|free-lsq}
                              #   --n-traj-fit N
                              #   --lsq-trunc-fs 500
  butane_kappa_convergence.py # kappa vs N subsample
  butane_gle_overleaf.tex     # this document
kernel_extraction/
  nonstationary_kernel_lsq.py  # Free-LSQ extractor
  benchmark_prony_nlsq.py     # Prony NLSQ extractor (tau_bounds option)
  benchmark_extended_cases.py # GLE integrators (multi-exp + arbitrary)

```

7 Open questions

- Compare the LSQ kernel against the equilibrium VolterraBasis kernel from the 40 ns unbiased run (plan 4, step 3).
- Position-dependent kernel $K(\phi_0, t)$: existing `PositionalKernel_fd.py` extracts $K(\phi, t)$ from the same shoots; not yet inserted into the GLE.
- Repeat the protocol from the trans well (plan 2–4) and check the implied gauche $\leftrightarrow$ trans rate against Eyring/Kramers.
- Decide whether the residual $0.241 \rightarrow 0.250$ gap is statistical (σ on κ_{GLE} from $N_{\text{traj}} = 5000$) or due to multidimensional reaction-coordinate effects.